

Probabilistic Graphical Models

CSE 5830

Representation

Lecture 5

CPD & Markov Network

Goals for Today

- Markov network (undirected graph)
 - Factors
 - Gibbs Distribution
 - Independence
 - active paths, separation
 - Representation theorems
 - Factorization and dependencies

Directional interaction may not be able to capture
all the independency

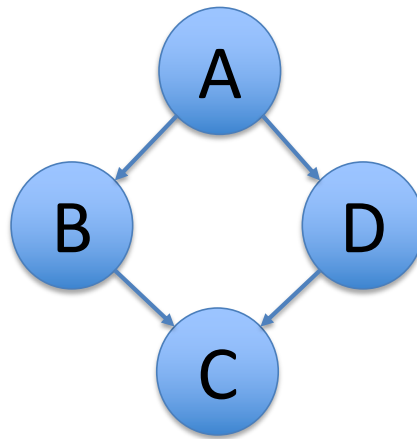
Example: Misunderstanding students

- $A \text{---} D, A \text{---} B$
- $C \text{---} D, C \text{---} B$
- $A \perp C | B, D$
- $B \perp D | A, C$

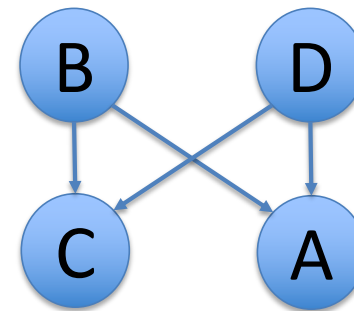
Directional interaction may not be able to capture all the independency

Misunderstanding students

- $A \perp C | B, D$
- $B \perp D | A, C$



Fail to capture $B \perp D | A, C$



Fail to capture $A \perp C | B, D$

Use undirected graph

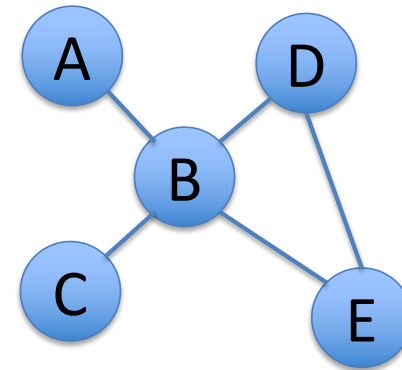
Undirected Graphical Models

- Also known as **Markov networks** and **Markov random fields**.
- Alternative representation of graphs with probability distributions.
 - MN can be viewed as encoding a set of independence assumption.

Markov Networks

Node, edge, and factor

- Each random variable is a node
- Undirected edges
- Factors are associated with edges or more generally subsets of nodes that form cliques.



Definition:

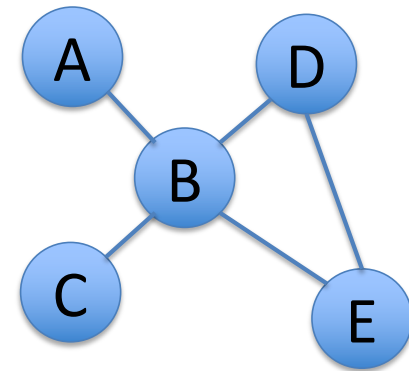
Clique: Fully connected subset of nodes.

Maximal Clique: Clique which is not a subset of a larger clique.

Factors in Markov Networks

- A factor maps assignments of its nodes to nonnegative “compatibility” values.
- General purpose function

$$\phi(A, B): Val(A, B) \rightarrow \mathbb{R}^+$$



- Affinity or compatibility function on edges or cliques

Example: Factors

Compatibility function or affinity function on edges

$$\phi_1(A, B)$$

a^0	b^0	30
a^0	b^1	5
a^1	b^0	1
a^1	b^1	10

$$\phi_2(B, C)$$

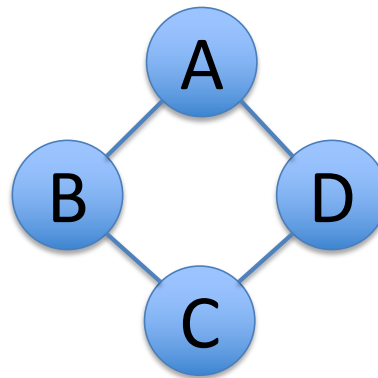
b^0	c^0	100
b^0	c^1	1
b^1	c^0	1
b^1	c^1	100

$$\phi_4(D, A)$$

d^0	a^0	100
d^0	a^1	1
d^1	a^0	1
d^1	a^1	100

$$\phi_3(C, D)$$

c^0	d^0	1
c^0	d^1	100
c^1	d^0	100
c^1	d^1	1



BN vs MRF

- Directed graphical models
 - Directed edges give causality relationships (Bayesian Network)
 - Parameterization $\leftrightarrow$ Independence
 - CPD
- Undirected graphical models
 - Undirected edges simply give correlations between variables (Markov Random Field)
 - Factors

How we can parameterize undirected graphs?

Factorization

- We can parameterize a graph by associating a set of factors with it
- Connection between factorization and independence properties

$$P \models (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) \quad \longleftrightarrow \quad P(\mathcal{X}) = \phi_1(\mathbf{X}, \mathbf{Z})\phi_2(\mathbf{Y}, \mathbf{Z})$$

Gibbs Distribution

- Set of factors: $\Phi = \{\phi_1(\mathbf{D}_1), \dots, \phi_K(\mathbf{D}_K)\}$
- Probability distribution is a **Gibbs** distribution:

$$P_{\Phi}(X_1, \dots, X_n) = \frac{1}{Z} \tilde{P}_{\Phi}(X_1, \dots, X_n),$$

Where

$$\tilde{P}_{\Phi}(X_1, \dots, X_n) = \phi_1(\mathbf{D}_1) \times \phi_2(\mathbf{D}_2) \times \dots \times \phi_m(\mathbf{D}_m)$$

Is an unnormalized measure and

$$Z = \sum_{X_1, \dots, X_n} \tilde{P}_{\Phi}(X_1, \dots, X_n)$$

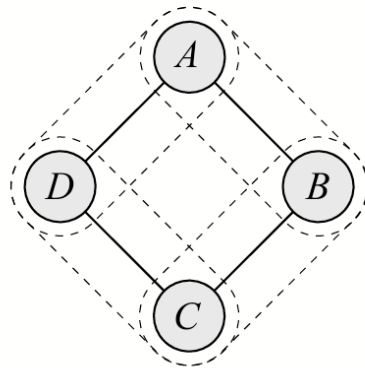
How to relate parameterization of a Gibbs distribution to a graph structure?

Markov network factorization

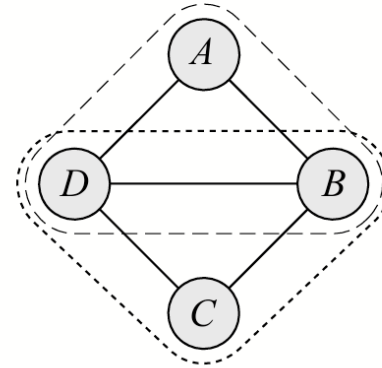
A distribution P_Φ with $\Phi = \{\phi_1(\mathbf{D}_1), \dots, \phi_K(\mathbf{D}_K)\}$ factorize over a Markov network H if each D_k ($k = 1, \dots, K$) is a complete subgraph of H

Complete subgraph $\rightarrow$ maximal cliques

$$\{\phi(A, B), \phi(B, C), \phi(C, D), \phi(D, A)\}$$



$$\{\phi(A, B, D), \phi(B, C, D)\}$$



Example: Probability distribution

- Gibbs distribution: Product of local factors

$$P(a, b, c, d) = \frac{1}{Z} \phi_1(a, b) \cdot \phi_2(b, c) \cdot \phi_3(c, d) \cdot \phi_4(d, a),$$

where

$$Z = \sum_{a,b,c,d} \phi_1(a, b) \cdot \phi_2(b, c) \cdot \phi_3(c, d) \cdot \phi_4(d, a)$$

 partition function

What is the probability of $a=1$, $b=1$, $c=0$, and $d=1$?

Example:

$$P(a^1, b^1, c^0, d^1) = \phi_1(a^1, b^1) \cdot \phi_2(b^1, c^0) \cdot \phi_3(c^0, d^1) \cdot \phi_4(d^1, a^1)$$

$$= 10 \cdot 1 \cdot 100 \cdot 100 = 100,000$$

$\phi_1(A, B)$			$\phi_4(D, A)$		
a^0	b^0	30	d^0	a^0	100
a^0	b^1	5	d^0	a^1	1
a^1	b^0	1	d^1	a^0	1
a^1	b^1	10	d^1	a^1	100

$\phi_2(B, C)$			$\phi_3(C, D)$		
b^0	c^0	100	c^0	d^0	1
b^0	c^1	1	c^0	d^1	100
b^1	c^0	1	c^1	d^0	100
b^1	c^1	100	c^1	d^1	1

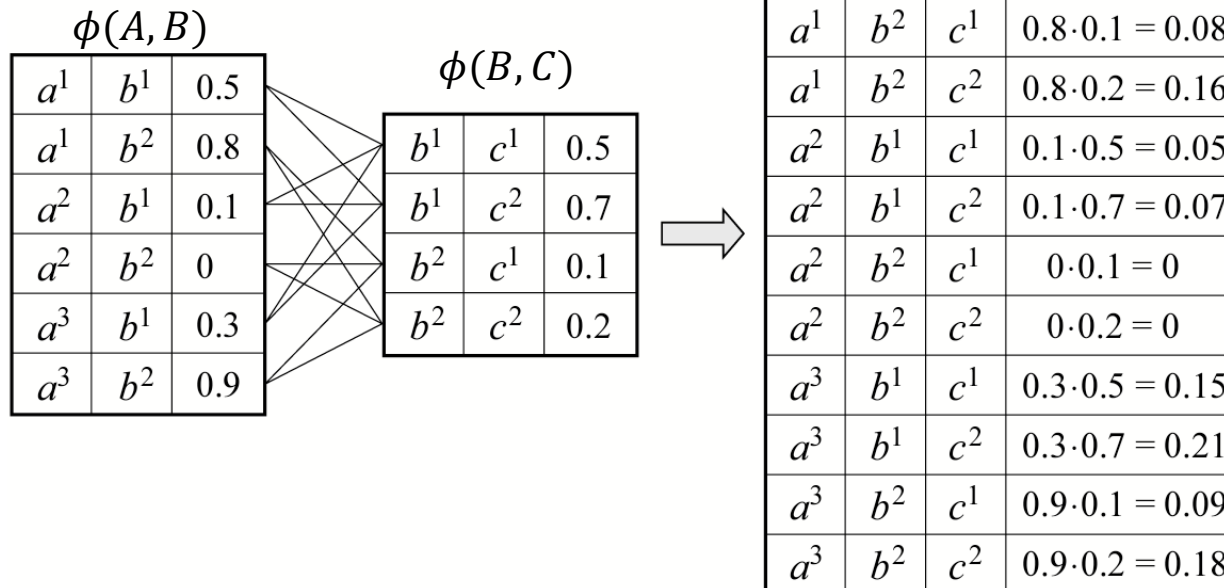
Assignment				Unnormalized	Normalized
a^0	b^0	c^0	d^0	300,000	0.04
a^0	b^0	c^0	d^1	300,000	0.04
a^0	b^0	c^1	d^0	300,000	0.04
a^0	b^0	c^1	d^1	30	$4.1 \cdot 10^{-6}$
a^0	b^1	c^0	d^0	500	$6.9 \cdot 10^{-5}$
a^0	b^1	c^0	d^1	500	$6.9 \cdot 10^{-5}$
a^0	b^1	c^1	d^0	5,000,000	0.69
a^0	b^1	c^1	d^1	500	$6.9 \cdot 10^{-5}$
a^1	b^0	c^0	d^0	100	$1.4 \cdot 10^{-5}$
a^1	b^0	c^0	d^1	1,000,000	0.14
a^1	b^0	c^1	d^0	100	$1.4 \cdot 10^{-5}$
a^1	b^0	c^1	d^1	100	$1.4 \cdot 10^{-5}$
a^1	b^1	c^0	d^0	10	$1.4 \cdot 10^{-6}$
a^1	b^1	c^0	d^1	100,000	0.014
a^1	b^1	c^1	d^0	100,000	0.014
a^1	b^1	c^1	d^1	100,000	0.014

Factors

- Fundamental building block for defining distributions in high-dimensional spaces
- Set of basic operations for manipulating these probability distributions
 - Factor product
 - Factor marginalization
 - Factor reduction

Factor product

$$\phi(A, B, C) = \phi(A, B) \times \phi(B, C)$$



Factor Marginalization

$\phi(A, B, C, D)$								
a^0	b^0	c^0	d^0	0.04		a^0	b^0	0.13
a^0	b^0	c^0	d^1	0.04				
a^0	b^0	c^1	d^0	0.04				
a^0	b^0	c^1	d^1	$4.1 \cdot 10^{-6}$				
a^0	b^1	c^0	d^0	$6.9 \cdot 10^{-5}$		a^0	b^1	0.69
a^0	b^1	c^0	d^1	$6.9 \cdot 10^{-5}$				
a^0	b^1	c^1	d^0	0.69				
a^0	b^1	c^1	d^1	$6.9 \cdot 10^{-5}$				
a^1	b^0	c^0	d^0	$1.4 \cdot 10^{-5}$		a^1	b^0	0.14
a^1	b^0	c^0	d^1	0.14				
a^1	b^0	c^1	d^0	$1.4 \cdot 10^{-5}$				
a^1	b^0	c^1	d^1	$1.4 \cdot 10^{-5}$				
a^1	b^1	c^0	d^0	$1.4 \cdot 10^{-6}$		a^1	b^1	0.04
a^1	b^1	c^0	d^1	0.014				
a^1	b^1	c^1	d^0	0.014				
a^1	b^1	c^1	d^1	0.014				

Factor reduction

$\phi(A, B, C)$

a^1	b^1	c^1	$0.5 \cdot 0.5 = 0.25$
a^1	b^1	c^2	$0.5 \cdot 0.7 = 0.35$
a^1	b^2	c^1	$0.8 \cdot 0.1 = 0.08$
a^1	b^2	c^2	$0.8 \cdot 0.2 = 0.16$
a^2	b^1	c^1	$0.1 \cdot 0.5 = 0.05$
a^2	b^1	c^2	$0.1 \cdot 0.7 = 0.07$
a^2	b^2	c^1	$0 \cdot 0.1 = 0$
a^2	b^2	c^2	$0 \cdot 0.2 = 0$
a^3	b^1	c^1	$0.3 \cdot 0.5 = 0.15$
a^3	b^1	c^2	$0.3 \cdot 0.7 = 0.21$
a^3	b^2	c^1	$0.9 \cdot 0.1 = 0.09$
a^3	b^2	c^2	$0.9 \cdot 0.2 = 0.18$

Reduce to the context $C=c^1$

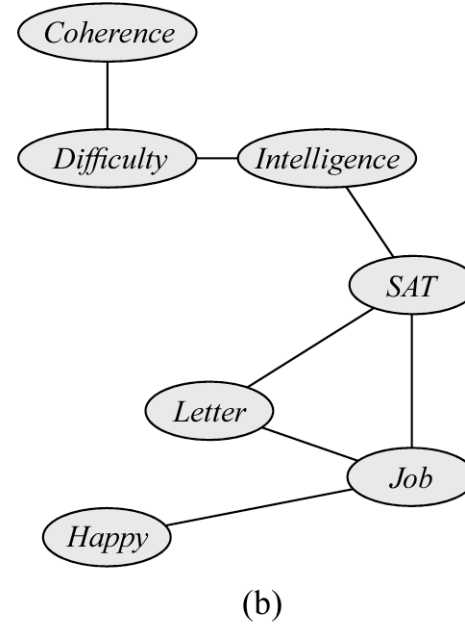
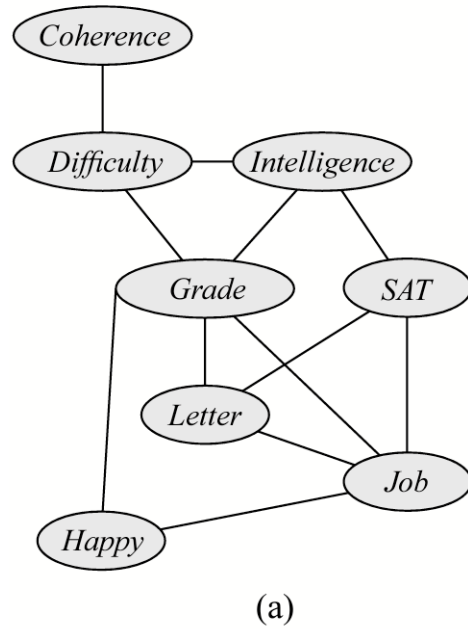
$\phi(A, B, C = c^1)$

a^1	b^1	c^1	0.25
a^1	b^2	c^1	0.08
a^2	b^1	c^1	0.05
a^2	b^2	c^1	0
a^3	b^1	c^1	0.15
a^3	b^2	c^1	0.09

What happen on the graph?

Factor reduction results a simpler graph

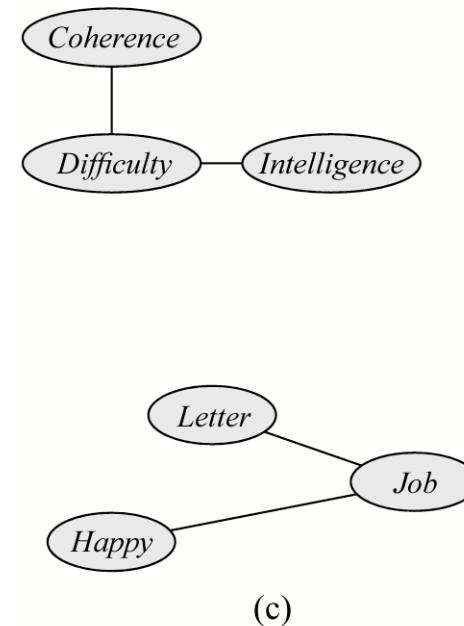
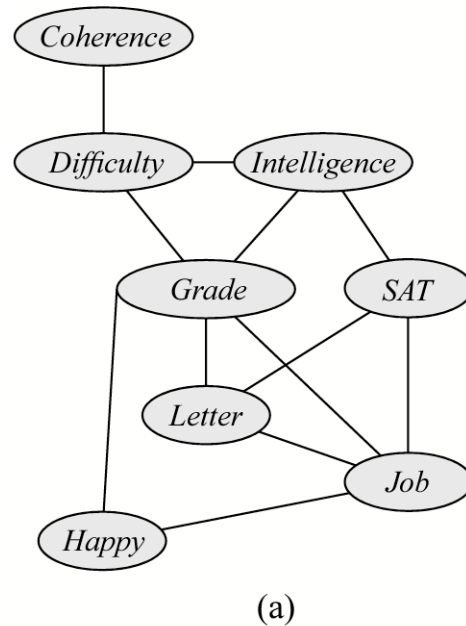
Example: Reduced to the context $G=g$



Simpler graph

Factor reduction results a simpler graph

Example: Reduced to the context $G=g, S=s$



Pairwise Markov network

- A subclass of Markov networks
- All factors associated with one node or one pair (connected by an edge).

$$P(\mathbf{X}) = \prod_i \phi_i(X_i) \prod_{(i,j) \in \mathcal{H}} \phi_{i,j}(X_i, X_j)$$

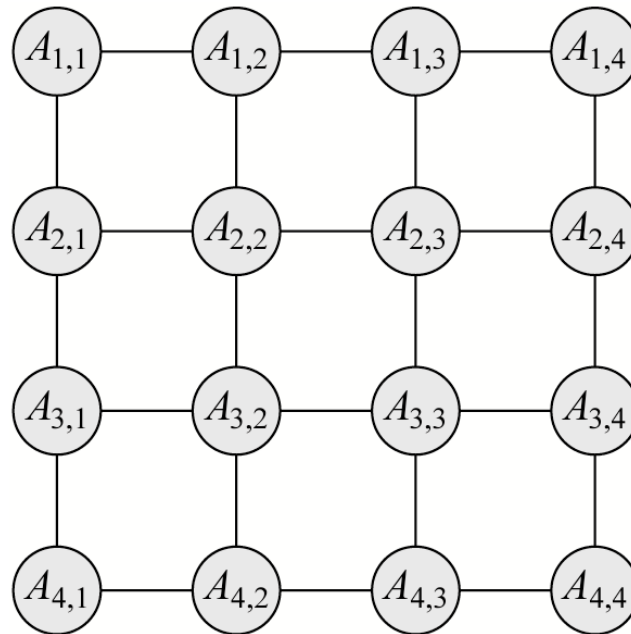
- The graph may have cliques with more than two nodes, but they do not have factors.

Pairwise Markov Network

A pairwise Markov network structured as a grid

A set of **node potentials**

A set of **edge potentials**



Pairwise Markov Network

Image denoising

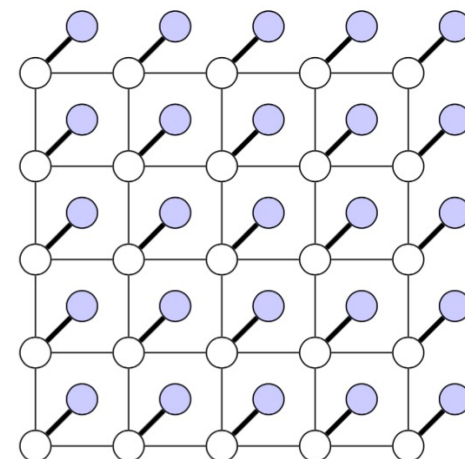
Problem: We want to recover a binary image from the observation of a corrupted version of it.

$X = \{X_i, i = 1, \dots, D\}$ $X_i \in \{-1, 1\}$: clean pixel

$Y = \{Y_i, i = 1, \dots, D\}$ $Y_i \in \{-1, 1\}$: corrupted pixel

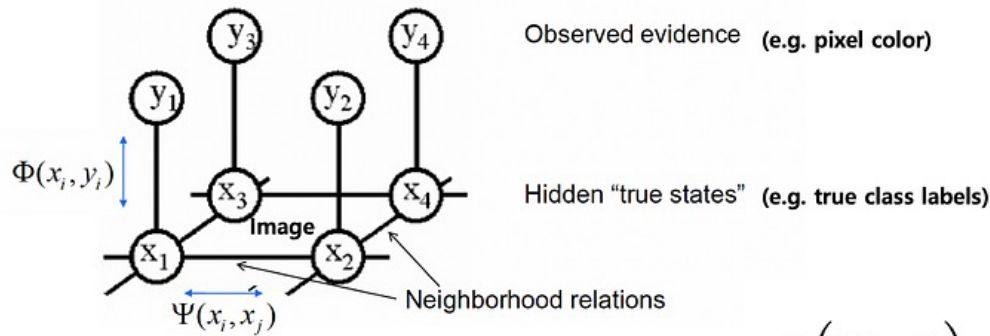
$\phi(Y_i, X_i) = e^{\gamma X_i Y_i}$ encourage Y_i and X_i to be similar

$\psi(X_i, X_j) = e^{\beta X_i X_j}$ encourage the image to be smooth



$$p(X, Y) \propto \left[\prod_{i=1}^D \phi(Y_i, X_i) \right] \left[\prod_{i \sim j} \psi(X_i, X_j) \right]$$

Pairwise Markov network image segmentation



$$p(\mathbf{x}, \mathbf{y}) = \prod_i \Phi(x_i, y_i) \prod_{i,j} \Psi(x_i, x_j)$$

Scene

Image

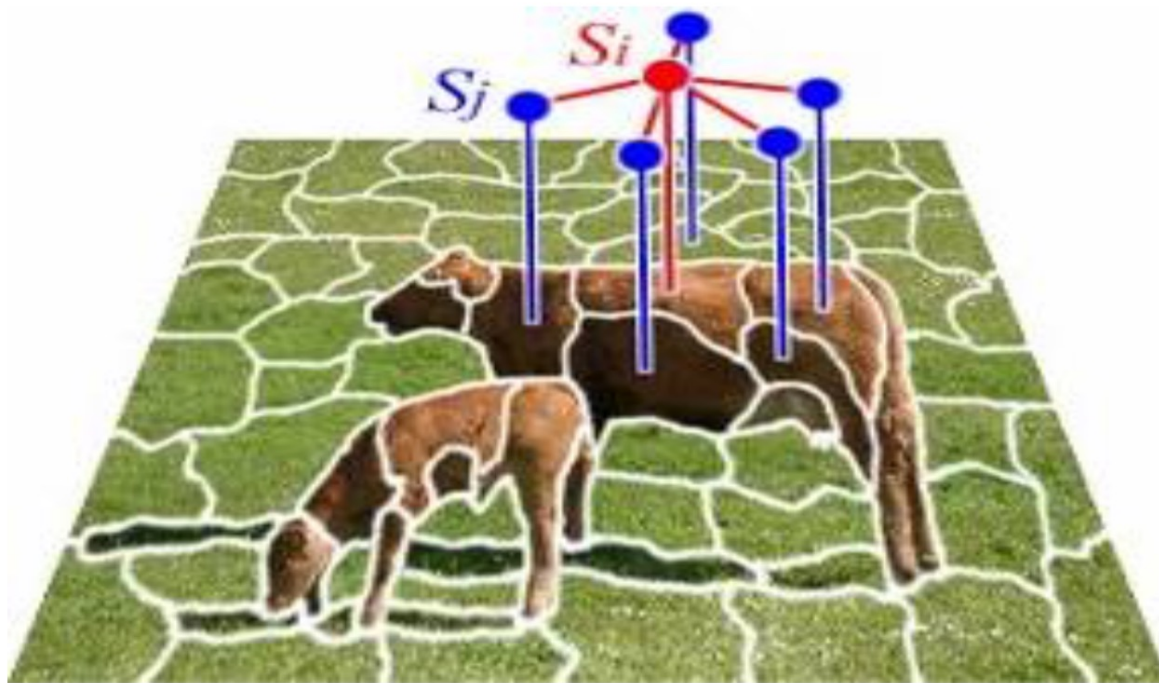
Image-scene compatibility function

Local observations

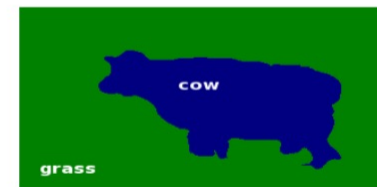
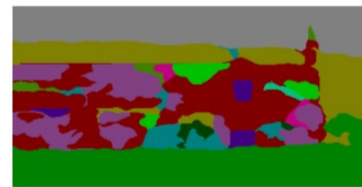
Scene-scene compatibility function

Neighboring scene nodes

Pairwise Markov network image segmentation



Pairwise Markov network image segmentation



(a)

(b)

(c)

(d)

Pairwise Markov network

Ising Models

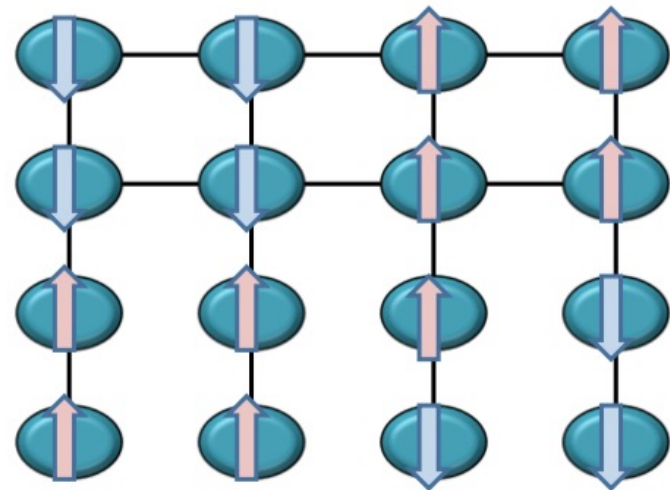
Ferromagnetism in statistical mechanics

$$X_i \in \{+1, -1\}$$

$$P(X) \propto \prod_{i < j} w_{i,j} x_i x_j \prod_i u_i x_i$$

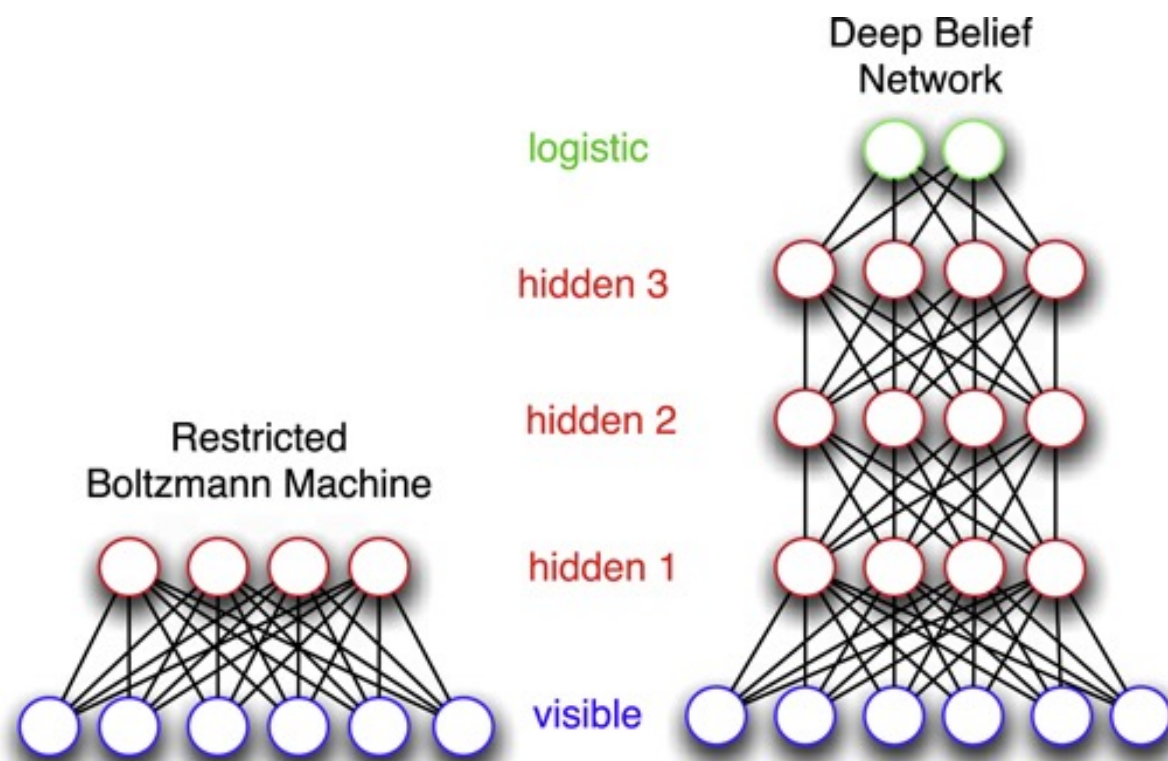
$$\phi_{i,j}(x_i, x_j) = w_{i,j} x_i x_j$$

$$\phi_i(x_i) = u_i x_i$$



Pairwise Markov network

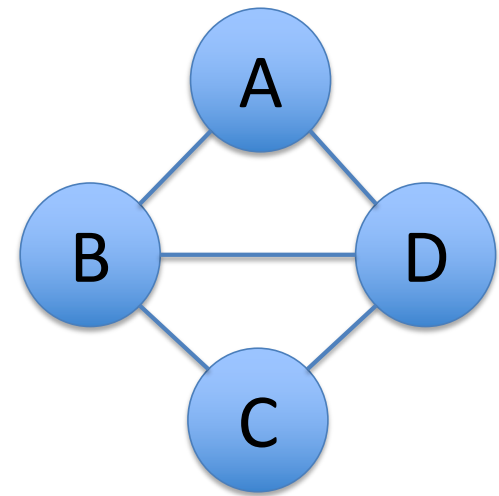
Boltzmann Machine and deep belief networks



How to relate a graph structure to factorization of a Gibbs distribution ?

- MN can be viewed as encoding a set of independence assumption
 - If we have the factorization we can induce a graph
- We cannot read factorization from graphs
 - different factorizations give the same graph
- Flow of Influence
 - Define independency
 - Influence can flow along any trail, regardless of the form of the factors

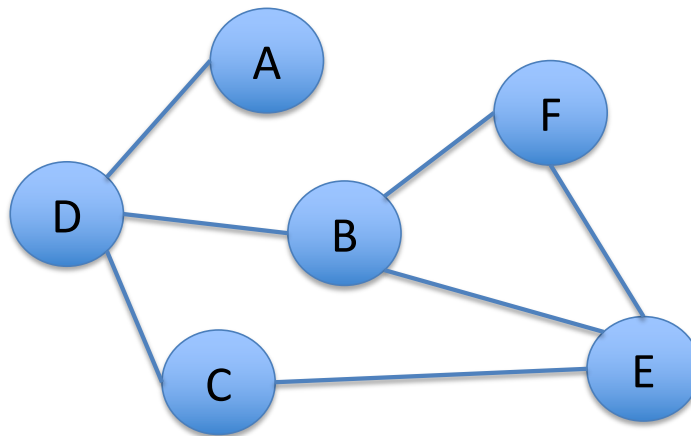
→ Active trail



- 1) $\phi(A, B), \phi(A, D), \phi(B, C), \phi(C, D), \phi(B, D)$
- 2) $\phi(A, B, D), \phi(B, D, C)$
- 3) $\phi(A, B, D), \phi(B, C), \phi(C, D)$

Active Trails

- A trail $X_1 - \dots - X_n$ is active given Z if no X_i is in Z
- Active trails depend only on graph structure



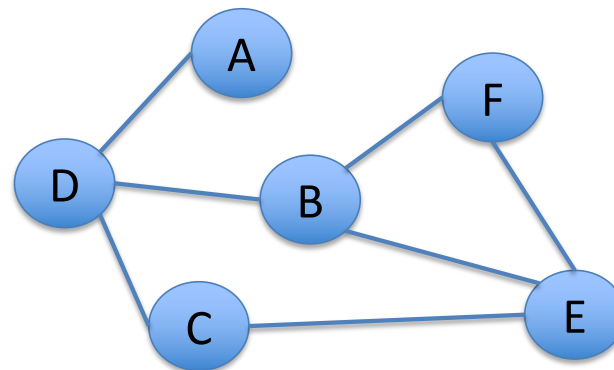
Independency in MN

- We can talk about flow of influence to define independency (Active trails)
- There are different types of independency
 - Global Markov dependency
 - Pairwise Markov dependency
 - Local Markov dependency

Global Markov independencies

- Separation in Markov networks
 - X and Y are separated in H given Z , $\text{sep}_{\mathcal{H}}(X; Y \mid Z)$ if there is no active trail in H between X and Y given Z
 - Global independencies associated with H :

$$\mathcal{I}(\mathcal{H}) = \{(X \perp Y \mid Z) : \text{sep}_{\mathcal{H}}(X; Y \mid Z)\}$$



Example: what we can say about A and E?

$A \perp E \mid D$?

$A \perp E \mid D, B$?

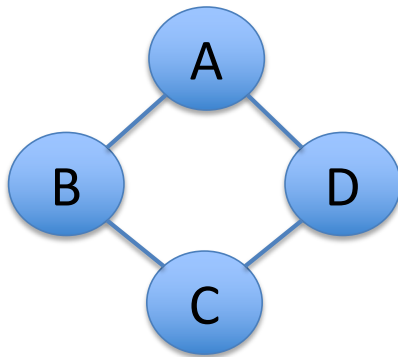
$A \perp E \mid B$?

$A \perp E \mid B, C$?

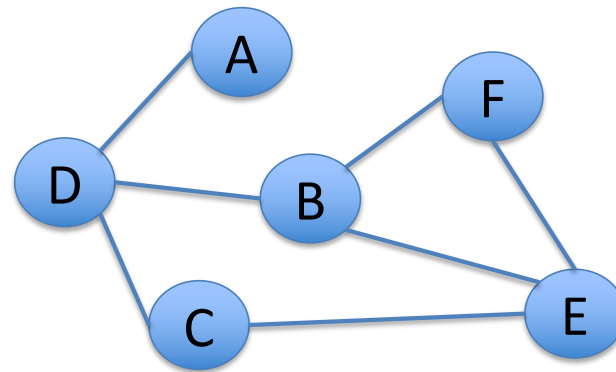
Pairwise Markov independencies

Pairs of *non-adjacent* variables are independent given everything else.

$$\mathcal{I}_p(\mathcal{H}) = \{(X \perp Y \mid \mathcal{X} - \{X, Y\}) : X - Y \notin \mathcal{H}\}$$



Example:
 $A \perp C \mid B, D$
 $B \perp D \mid A, C$
 $A \perp D \mid B, C$?

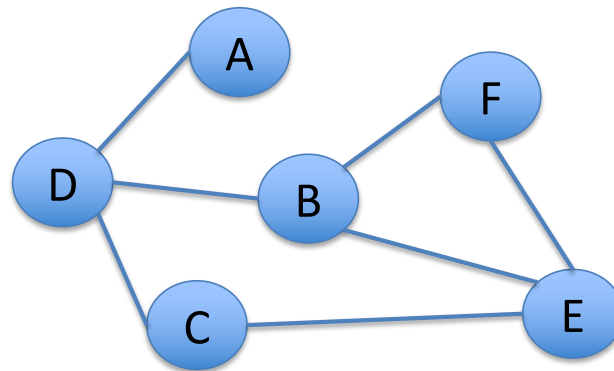


Example:
 $B \perp A \mid D, E, F, C$
 $C \perp A \mid D, B, F, E$
 $F \perp A \mid D, C, B, E$

Local Markov independencies

Markov blanket ($\text{MB}_{\mathcal{H}}(X)$): each variable is independent of the rest given its neighbors.

$$\mathcal{I}_{\ell}(\mathcal{H}) = \{(X \perp \mathcal{X} - \{X\} - \text{MB}_{\mathcal{H}}(X) \mid \text{MB}_{\mathcal{H}}(X)) : X \in \mathcal{X}\}$$



Example:

$B \perp A, C \mid D, E, F$

$C \perp A, B, F \mid D$

$F \perp A, D, C \mid B, E$

Summary Independence assumptions in MNs

- Global independencies, $I(H)$, separation:
$$\text{sep}_H(X, Y \mid Z) \Rightarrow X \perp Y \mid Z$$
- Pairwise independencies, $I_p(H)$:
$$A \perp B \mid X \setminus \{A, B\}$$
- Local independencies, Markov blanket, $I_l(H)$:
$$A \perp X \setminus \text{Neighbors}(A) \mid \text{Neighbors}(A)$$

$I_p(H)$ is strictly weaker than $I_l(H)$, which is strictly weaker than $I(H)$

- If $I_l(H) \rightarrow I_p(H)$
- If $I(H) \rightarrow I_l(H)$

Independency and BNs

Representation, revisited

- Bayesian networks ...

The Bayesian network graph is an I-map for P.



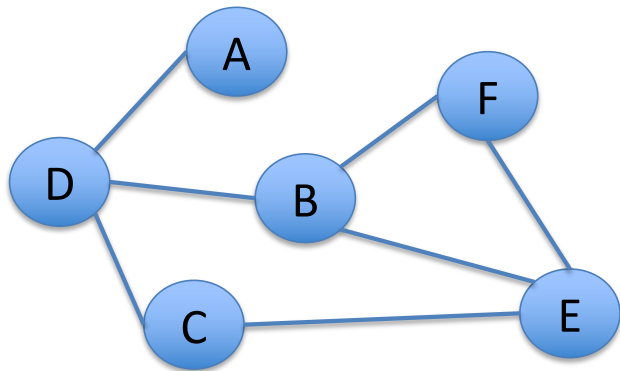
$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i \mid \mathbf{Parents}(X_i))$$

- Independencies give you the Bayesian network.
- Bayesian network reveals independencies.

Is it true for MNs?

Factorization $\Rightarrow$ Independence

Theorem: If P factorizes over H , and $sep_H(X, Y \mid Z)$ then P satisfies $(X \perp Y \mid Z)$



A, E are separated given D $\rightarrow$

P satisfies: $A \perp E \mid D$

A, E are separated given D, B $\rightarrow$

P satisfies: $A \perp E \mid D, B$

A, E are separated given B, C $\rightarrow$

P satisfies: $A \perp E \mid B, C$

Factorization $\Rightarrow$ Independence I-map

$$I(H) = \{(X \perp Y | Z) : sep_H(X, Y | Z)\}$$

- If P satisfies $I(H)$, we say that H is an I-map (independency map) of P

Theorem: If P factorizes over H , then H is an I-map of P

Independence $\Rightarrow$ Factorization

- **Theorem** (Hammersley Clifford):

For a positive distribution P , if H is an I-map for P , then P factorizes over H

Independency and MNs Representation

Markov networks ...

The Markov network's
graph is an I-map for P.



$$P(\mathbf{X}) = \frac{1}{Z} \prod_{i=1}^m \phi_i(\mathbf{D}_i)$$

From distribution to graph

Can we construct a graph for a distribution?

$$P \propto \prod_i \phi_i(\mathbf{X}_i) \rightarrow \text{build a graph } H$$

- Two equivalent views of graph structure:
 - Factorization: H allows P to be represented
 - I-map: Independencies encoded by H hold in P
- If P factorizes over a graph H , we can read from the graph independencies that must hold in P (an independency map)

Capturing independencies

- P factorizes over H  H is an I-map for P
- There can be independencies in $I(P)$ that are not in $I(H)$
 - $I(P)$: lists all independency statements in P
 - $I(H)$: for graph H , lists all of possible independencies
 -

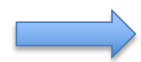
$$I(P) = \{(X \perp Y | Z) : P \models (X \perp Y | Z)\}$$

$$I(H) \subseteq I(P)$$

We want to capture all independencies, if possible

Minimal I-map

- The graph encodes more independencies
 - sparser (has fewer parameters)
 - more informative
- Want a graph that captures as much of the structure in P as possible



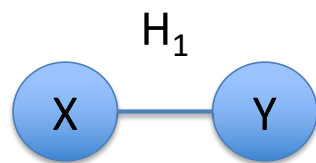
minimal I-map for P

- Minimal I-map: I-map without redundant edges
- Minimal I-map may still not capture $I(P)$

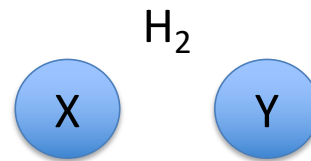
Minimal I-map

- A graph H is a minimal I-map for distribution P if
 - H is an I-map for P
 - Removing any edge from H , renders H not an I-map for P

Example: $P(Y|X = 0) = P(Y|X = 1)$



H_1 : I-map for P



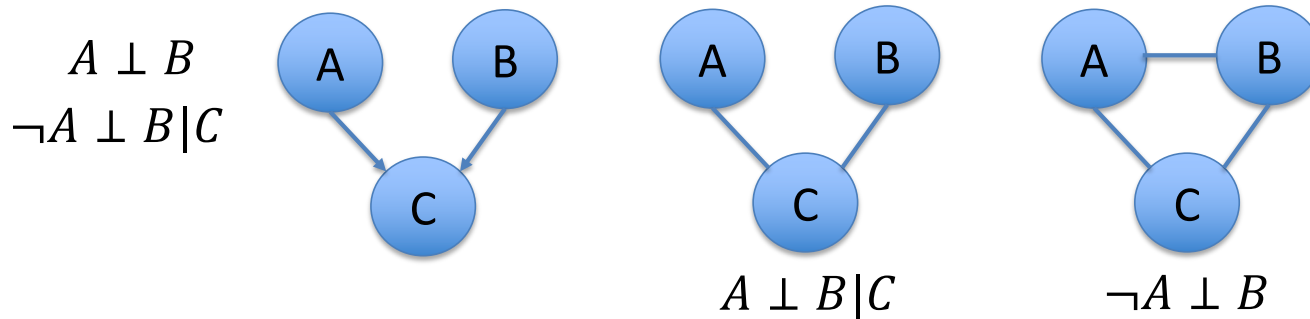
H_2 : minimal I-map for P

Perfect map (P-map)

- Want: independencies from the graphical model are exactly the same as in P.

➡ Perfect map: $I(H) = I(P)$

- Doesn't always exist for Bayesian networks (misunderstanding students).
- Doesn't always exist for Markov networks.



From distribution to graph

- Encode independencies in P using a graph H
- Build a minimal I-map for P
 - Pairwise Markov independencies
 - Introduce an edge X - Y for all X, Y
 - Local independencies, Markov blanket
 - Introduce an edge X - Y for all X and all Y in $MB_p(X)$

There might not be a unique H as a minimal I-map for P

I-equivalence

- Two graphs G_1 and G_2 over $X_1, \dots, X_n$ are I-equivalent if $I(G_1) = I(G_2)$
- Most graphs have many I-equivalent variants

